

Event-Triggered Communications Over AWGN Channels

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Abstract—Event-triggered communication (ETC) has attracted considerable recent attention because it holds the promise of saving bandwidth and energy in real-time monitoring, networked control, distributed optimization, federated learning, etc. However, its performance limits remain open from a communication-theoretic perspective. This paper investigates event-triggered communications over AWGN channels, where events are modeled by a semi-Markov process. An ETC-oriented code is conceived by adding an all-zero codeword that represents a non-triggered state without any additional energy cost to the random codebook. We present the maximum achievable channel coding rates in both finite and infinite blocklength regimes. Even though the newly added codeword may incur a higher error probability, the rate of infinite blocklength coding is found to remain unchanged, thereby assuring the applicability of Shannon’s formula in ETC. Furthermore, we present two multi-user ETC protocols based on statistical multiplexing and joint coding respectively. Borrowing the idea of information freshness and queueing analysis, we reveal the distortion-power tradeoffs of single- and multi-user ETC. The ETC with Lebesgue sampling is shown to outperform the conventional periodic communication with Riemann sampling. The number of users in statistical multiplexing is also optimized. Numerical results validate our theoretical analysis and demonstrate the substantial power-saving gain of ETC.

Index Terms—Event-triggered communications, networked control, distributed learning, Lebesgue sampling, channel capacity with cost constraint, coding for noisy channels, Shannon’s formula, distortion, queueing, information freshness, mean square error, statistical multiplexing, joint coding, performance analysis.

I. INTRODUCTION

IN THE upcoming era of 6G, stringent requirements for high energy efficiency and low-latency control have emerged [1]. Among various research areas, event-triggered communication (abbreviated as ETC in the rest of this paper) shows the potential for energy savings in areas such as industrial Internet of things (IIoT), smart grids, and intelligent transportation [2].

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In particular, we focus on the real-time monitoring and control of dynamic systems.

Dynamic systems traditionally employ a periodic, or time-triggered, sampling scheme. In order to ensure optimal performance, however, small sampling intervals are often used, which may lead to congestion due to redundant data transmission. To address this challenge, especially for scenarios with limited communication resources, ETC is proposed as an alternative. In [3], the performance of two different sampling strategies was compared under first-order systems. By using an analog of integral theory, periodic sampling is also called Riemann sampling, while event-based sampling is also called Lebesgue sampling. The results show that Lebesgue sampling gives substantially smaller variances for the same average sampling rates.

In a class of straightforward event-triggered strategies, a sampling or transmission is triggered whenever the signal exceeds a threshold or one of the preset levels. This is called level cross-sampling in signal processing, and the idea is similar to analog-to-digital conversion, data compression [4], and delta modulation [5]. Various applications of the level-crossing idea include bandwidth estimation [6], compressive sensing [7], and feedback control [8]. Since we can convert stochastic processes with continuous amplitude into transitions between finite states through methods such as level-cross sampling, our model can be abstracted as a “discrete-event system (DES)”. DES was introduced in the early 1980s to identify a class of dynamic systems whose behavior is governed by discrete events occurring asynchronously over time and solely responsible for generating state transitions [9]. In between event occurrences, the state of such systems remains unchanged. The control of DES is studied in [10].

Turning our attention to general continuous-time dynamic systems, research in the field of ETC mainly focuses on linear systems [11]. In [12], a periodic ETC (PETC) strategy is introduced, aiming to strike a balance between conventional periodic sampled-data control and ETC. In PETC, event-triggering conditions are verified periodically, and a decision is made at each sampling time whether to transmit new data. In [13], an event-based control approach for linear systems with disturbances is proposed to ensure exponential practical stabilization. More recently, in [14] and [15], ETC for multi-agent average consensus was studied, which can be applied to specific stochastic process tracking problems as well.

In addition to the related works above in the field of automatic control, the idea of ETC is also used in issues such as remote estimation and real-time monitoring [16]. In [17],

the remote estimation with communication costs for first-order linear systems is studied. At every time step, a real number with a positive cost or a free symbol with zero cost is transmitted to the remote estimator, which can be regarded as an early implementation of PETC. In [18], researchers specifically introduced the concept of implicit information: the time intervals between events imply that the signal remains between neighbouring levels. The role of implicit information is further studied in [19] and [20] and is referred to as timing side information (TSI).

In the framework of status updates, the age of information (AoI) has been introduced in queueing systems to capture the notion of information freshness, and many works on AoI/AoII-aware sampling can also be classified as ETC. In [21], however, it is shown that the optimal minimum MSE policy of a Wiener process can be much better than the AoI-optimal strategy. In [22], a new metric called the age of incorrect information (AoII) is introduced, and the problem of monitoring the states of remote sources that evolve according to a Markov process is studied. Although AoII addresses the shortcomings of earlier metrics such as AoI and the age of synchronization (AoS), it is still not equivalent to the metric of time-averaged MSE. To further clarify the novelty of our work, we review a series of relevant studies on AoI/AoII-aware sampling, including [22], [23], [24], [25], [26] as representative examples. A common feature of these works is that they measure communication cost by average transmission frequency. In other words, these studies do not account for the additional costs associated with non-uniform transmission in time. This leads to two key differences between our work and other related studies.

First, we consider how the receiver distinguishes between the transmitter's active (triggered) and idle states from a physical layer perspective. In contrast, most research on sampling strategies approaches this from the application layer. Consequently, our ETC framework can be used alongside AoII-aware sampling strategies to enhance system performance. Second, we use average power, rather than average transmission frequency, as the metric for communication cost, which provides a more equitable comparison of ETC and periodic communication. According to Shannon's capacity formula and Jensen's inequality, periodic transmission achieves a higher average rate than burst transmission for a given bandwidth and average power. Therefore, using average transmission frequency as the cost metric would create an unfair comparison.

Next, we specify our focus on implementing ETC from the physical layer. Unlike traditional periodic communication, ETC involves two states:

- Triggered state: Whenever a specific event occurs (e.g., the system state changes), a packet is generated based on the sample and awaits transmission. The triggered state is represented by $\mathcal{D}$.
- Idle state: In the absence of triggering events, the transmitter remains idle and consumes no power. The idle state is represented by $\bar{\mathcal{D}}$.

The transmitter switches between these states, raising the question of whether the receiver can distinguish the transmitter's status effectively. Dolk et al. considered specific

medium access protocols in [27]. Our approach treats the idle transmitter as sending an equivalent codeword θ and analyzes the block error rate on the expanded codebook.

In this way, the error rate of determining the transmitter status can be calculated by the decoding error between the all-zero codeword and the other regular codewords.

This necessitates research on finite blocklength coding and block error rates. Shannon initially provided tight bounds on error rates for the additive white Gaussian noise (AWGN) channel in his seminal work [28]. Since then, numerous studies have expanded upon these results in the context of finite blocklength analysis. Polyanskiy et al. extended the results to a broader class of channels, establishing the relationship between channel capacity and channel dispersion [29]. More recently, Collins and Polyanskiy explored dispersion and coding scheme design for coherent block-fading multi-antenna channels [30], while Lancho et al. examined the performance of finite blocklength coding in noncoherent single-antenna Rayleigh fading channels [31]. Li's recent work delved into the trade-offs between reliability, latency, and coding rate [32]. However, these studies primarily focus on periodic transmission. In contrast, our work investigates the performance of finite blocklength coding in ETC scenarios, considering the impact of non-uniform transmission over time. This perspective, particularly from the physical layer, distinguishes our approach from previous studies.

In this paper, we comprehensively explore the feasibility and performance of ETC. Our contributions are summarized as follows:

- 1) We analyze the additional communication cost of ETC compared to traditional periodic communication from an information-theoretic perspective. In the infinite blocklength regime, we show that the AWGN channel capacity conditioned on the triggered state in ETC is consistent with Shannon's formula. In the finite blocklength regime, we demonstrate a reduction in channel coding rate by presenting the relationship between the channel coding rate, block error rate, and blocklength. These results are then extended to fading channels and other more general channel models.
- 2) We analyze the distortion-power tradeoff of ETC for a special class of semi-Markov processes. Instead of delay-related metrics such as AoII, we assess real-time reconstruction distortion by using the misalignment time ratio as an equivalent metric as well as an intermediate step. This provides a fair comparison of performance evaluation between ETC and periodic communication in real-time monitoring systems.
- 3) The performance analysis extends to multi-user scenarios, where we investigate two strategies: statistical multiplexing and joint coding. For statistical multiplexing-based ETC, we model the system as a G/D/1 queue and derive the average distortion using tools such as the generating function of the stationary distribution and the finite-source queue model. For joint coding, we evaluate the distortion-power tradeoff by calculating the entropy of joint samples and applying results from ordered statistics.

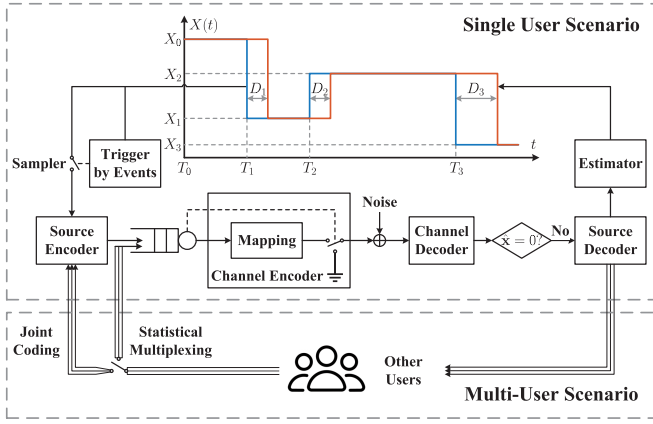


Fig. 1. System model.

- 4) We examine the optimal number of users for a multi-user shared channel, providing insights into maximizing system performance under varying conditions. This analysis offers practical guidelines for resource allocation and scheduling in shared communication environments.

The remainder of this paper is organized as follows. Section II introduces the system model, the problem formulation, and the characteristics of event-triggered channels. Section III analyzes the theoretical channel coding rate in finite and infinite blocklength regimes. Section IV focuses on the average distortion and power analysis of ETC for a special type of jumping process, considering the strategies based on statistical multiplexing and joint coding respectively. Section V presents the results of numerical simulations. Finally, Section VI summarizes the main conclusions of this paper.

II. SYSTEM MODEL

As shown in Fig. 1, we consider a remote monitoring system consisting of a source, a channel, and an estimator. The source generates continuous time signals $X(t)$ at time t representing the state of a physical process or a sensor measurement. The source process is sampled, encoded, and modulated before being sent over the channel. In scenarios with limited communication resources, each transmission consumes a certain cost and introduces transmission delay, thereby introducing distortion between the remote real-time estimated signal $\hat{X}(t)$ and the source signal $X(t)$. We use p to denote the probability that the transmitter is triggered. Our goal is to minimize the estimation error with limited communication costs. In addition, we focus on the performance improvement of sharing communication resources among multiple users.

A. Source Model

In this paper, the monitored source $X(t)$ is characterized by a semi-Markov process, which transitions after a certain amount of holding time τ . The jump process $X(t)$ takes states in the set $S = \{\alpha_1, \alpha_2, \dots, \alpha_{|S|}\}$ and is described by a set of random variables (X_m, T_m) , where T_m represents the jump times and X_m represents the sequence of source states.

Therefore, the source process can be expressed as $X(t) = X_m$ for $t \in [T_m, T_{m+1})$, and the sequence of holding times is given by $\tau_m = T_m - T_{m-1}$, where $T_0 = 0$. For the convenience of analysis, we assume that X_m is a sequence of independently and identically distributed (i.i.d.) random variables with a distribution of $\Pr\{X = \alpha_i\} = p_i$.

A semi-Markov process is chosen for its generality and suitability in modeling ETC. In event-triggered systems, events occur at discrete times with random intervals. Each event can be of a different type, requiring state descriptions to capture their nature. Semi-Markov processes accommodate these characteristics by allowing state transitions after random holding times, with each state representing a different event type. This flexibility makes the semi-Markov process an effective framework for analyzing ETC, aligning well with the random and discrete nature of event occurrences in such systems.

B. AWGN Channels

In this subsection, we specify the AWGN channel and the cost of its symbol transmission. In channel coding with cost constraints, we take into account the cost of each codeword as in [33, Chapter 3.6]. Let $\mathcal{X}$ denote the input alphabet of the channel. For $n = 1, 2, \dots$, define mapping $c_n : \mathcal{X}^n \rightarrow \mathbb{R}$, where $\mathbb{R}$ denotes the set of all real numbers. For $\mathbf{x} \in \mathcal{X}^n$ we call $c_n(\mathbf{x})$ the cost of $\mathbf{x}$ and $\frac{1}{n}c_n(\mathbf{x})$ the cost per symbol of $\mathbf{x}$. In particular, we use θ to represent the equivalent input symbol of the channel in the idle state. For ETC, it is assumed that the cost in the idle state is 0, i.e., $c(\theta) = 0$.

In this paper, we focus on the AWGN channel, where the codeword cost is equal to its power, i.e., $c_n(\mathbf{x}) = \sum_{i=1}^n x_i^2$. The idle state equivalent codeword is $\theta = 0$. The equivalent baseband real-valued channel model can be expressed as

$$\mathbf{y} = \mathbf{x} + \mathbf{z}, \quad \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n \quad (1)$$

where $\mathbf{z}$ denotes the additive noise. Without loss of generality, we normalize the variance of the noise, such that $\mathbf{z}$ follows a Gaussian distribution $\mathcal{N}(\mathbf{0}, N\mathbf{I}_n)$. Then $\{z_i\}_{i=1}^n$ is a set of i.i.d. normal random variables with variance N .

The transmitted signal $\mathbf{x}$ is selected from the codebook $\mathcal{C}_n = \{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_M\}$, where the cardinality of the codebook is $|\mathcal{C}_n| = M$ and each codeword $\mathbf{s}_i \in \mathbb{R}^n$. Thus, the coding rate (bit per dimension) can be expressed as $R = \frac{1}{n} \log M$. Unless otherwise specified, the “log” used throughout this paper denotes the base 2 logarithm. Let P denote the signal power constraint and $\gamma = P/N$ represent the signal-to-noise ratio (SNR). There are typically three types of power constraints:

- equal-power constraint: each codeword $\mathbf{s}_i \in \mathbb{R}^n$ satisfies $|\mathbf{s}_i|^2 = nP$.
- maximal power constraint: each codeword $\mathbf{s}_i \in \mathbb{R}^n$ satisfies $|\mathbf{s}_i|^2 \leq nP$.
- average power constraint: all codewords in the codebook satisfy $\frac{1}{M} \sum_{i=1}^M |\mathbf{s}_i|^2 \leq nP$.

These three power constraints have decreasing stringency. For example, if a set of codewords satisfies the equal-power constraint, then it naturally satisfies the maximal power constraint and the average power constraint. It should be noted that all three constraints lead to quite similar results [28].

C. Performance Metrics

We first define online sampling policies, in which each sampling time χ_i is selected causally. Specifically, the sampler only uses currently available information. To characterize this statement, we define

$$\mathcal{F}_t = \sigma(X(s) : 0 \leq s \leq t), \quad \mathcal{F}_t^+ = \cap_{s>t} \mathcal{F}_s, \quad (2)$$

where $\sigma(Y_1, Y_2, \dots, Y_n)$ denotes the σ -field generated by the random variables $Y_1, Y_2, \dots, Y_n$. Accordingly, $\mathbb{F}^+ = \{\mathcal{F}_t^+, t \geq 0\}$ represents a right-continuous filtration, which is a family of σ -fields ordered non-decreasingly, containing the available information at the observer. Each sampling time χ_i is a stopping time with respect to the filtration $\mathbb{F}^+$, i.e.,

$$\{\chi_i \leq t\} \in \mathcal{F}_t^+, \quad \forall t \geq 0. \quad (3)$$

For example, for periodic sampling, we have $\chi_{i+1} = \chi_i + T_s$, where T_s denotes the sampling period. For event-triggered sampling, the sampling times are given by $\chi_{i+1} = \inf\{t > \chi_i : X(t) \neq X(\chi_i)\}$. According to the causality, the available information to the receiver at time t is given by

$$\mathcal{G}_t = \{X(\chi_i) \mid \chi_i + D_i < t, \quad i = 1, 2, 3, \dots\}, \quad (4)$$

where D_m denotes the delay of the packet corresponding to the i -th sample. Assume that the channel transmits the packets in a first-in, first-out order, so that the order of the received packets is the same as the order of transmission, i.e., $\chi_i + D_i < \chi_j + D_j$ for all $i < j$. Let $\mathcal{U}$ denote the decoding function. The reconstructed signal is given by

$$\hat{X}(t) = \mathcal{U}(\mathcal{G}_t) \quad (5)$$

We evaluate system performance based on two key aspects: communication resource constraints and average distortion. For communication resource constraints, we consider P_{ave} as the average transmit power per user and W as the bandwidth per user. Regarding average distortion, we define a Borel measurable function $\psi : \mathbb{R} \times \mathbb{R} \rightarrow [0, \infty)$ for each t . The distortion at time t is measured by

$$J(t) = \mathbb{E} \left[\psi \left(X(t), \hat{X}(t) \right) \right]. \quad (6)$$

The average distortion is $J = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T J(t) dt$. In this paper, we adopt the mean square error (MSE) as the measure ψ . So the time-averaged distortion is given by

$$J = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mathbb{E} \left[\left(X(t) - \hat{X}(t) \right)^2 \right] dt. \quad (7)$$

Note that J is a deterministic function of the sampling and transmission strategy due to the ergodic nature of the source process.

Next, we define another performance metric, the misalignment rate η , which represents the proportion of time that the receiver fails to keep up with the source state updates. Specifically, the receiver is misaligned with its source whenever a newly generated packet has not been fully transmitted. If the reconstructed signal is determined by the last received sample, i.e.,

$$\hat{X}(t) = X(\chi_i), \quad \text{if } t \in [\chi_i + D_i, \chi_{i+1} + D_{i+1}). \quad (8)$$

we can establish the following proposition:

Proposition 1: The average distortion J is proportional to the misalignment ratio η and is given by

$$J = \eta \Theta, \quad (9)$$

where Θ denotes the average distortion under the random reconstruction strategy and is given by

$$\Theta = \sum_{i=1}^{|S|} \sum_{j=1}^{|S|} p_i p_j (\alpha_i - \alpha_j)^2. \quad (10)$$

III. CHANNEL CODING OF ETC

This section explores the channel coding of ETC from an information-theoretic perspective. In conventional communication, the transmitter samples the process periodically, encoding each sample into a codeword from a random codebook, thereby consuming some power [34, Chapter 9]. In contrast, when ETC is adopted, a codeword is sent only if an event triggers transmission; otherwise, the transmitter remains idle without power consumption. We first define the ETC-specific codebook and highlight its differences from traditional periodic communication. Then, we analyze the channel coding rate of ETC in both finite and infinite blocklength regimes.

A. Channel Capacity in the Infinite Blocklength Regime

In this subsection, we analyze the channel capacity of ETC in the infinite blocklength regime. We begin by describing a scenario well-suited for ETC. In this scenario, a continuous-time semi-Markov process is periodically sampled, and a symbol sequence $\mathbf{x}_n$ is generated by comparing the current sampling result with the previous one. The sampling results are then mapped to an extended codebook $\mathcal{C}'_n$ according to the following rule:

$$x_k = \begin{cases} \mathbf{s}_i, & X(kT_s) \neq X((k-1)T_s) \\ \theta, & X(kT_s) = X((k-1)T_s), \end{cases} \quad (11)$$

where θ represents the idle state equivalent symbol. For AWGN channels, an additional all-zero symbol $\theta = \mathbf{0}$ is added to the original codebook, so the equivalent codebook for ETC is $\mathcal{C}'_n = \mathcal{C}_n \cup \mathbf{0}$. Since the probability of the triggered state is p , the probability of the codeword θ is $1 - p$, and the probability of the codeword $\mathbf{s}_i$ is p/M for all $i = 1, 2, \dots, M$.

The following theorem presents the channel capacity of ETC over AWGN channels. The achievability of this capacity is demonstrated using a constructive method, which involves using a random codebook and a two-step decoding scheme. In the first step, the trigger state of the transmitter is determined by comparing the signal amplitude to a predefined threshold. In the second step, once the trigger state is identified, the specific codeword is selected using the maximum likelihood criterion. The converse of the capacity is comparatively easier to prove.

Theorem 1: For ETC over an AWGN channel with SNR γ , the channel capacity in the triggered state is the same as that in conventional communication and is given by

$$C_{\text{ETC}|\mathcal{D}} = \frac{1}{2} \log(1 + \gamma). \quad (12)$$

Proof: See Appendix. ■

Theorem 1 establishes that the AWGN channel capacity in the triggered state under ETC is equivalent to the capacity in conventional communication, even with the added requirement of distinguishing additional transmitter states. This result implies that, in terms of channel capacity, no additional communication resources are necessary to ensure the reliability of ETC. We thus prove that various strategies with non-uniform transmission in time are theoretically feasible at the physical layer.

B. Error Rate in the Finite Blocklength Regime

In this subsection, we analyze the error rate of ETC in the finite blocklength regime. Upper and lower bounds are derived by geometrically interpreting and scaling the decoding error region for received symbols. We adopt the maximum a posteriori (MAP) criterion based on the extended codebook $\mathcal{C}'_n$. As illustrated in Fig. 2a, the decision boundary between $\mathbf{s}_i$ and $\mathbf{s}_j$ ($i \neq j$) is their perpendicular bisector. For θ and $\mathbf{s}_i$ with different prior probability, We define r_0 and r_1 as

$$r_0 = \frac{\sqrt{n\gamma}}{2} \left(1 + \frac{2R \ln 2}{\gamma} + \frac{2 \ln \frac{1-p}{p}}{n\gamma} \right), \quad (13)$$

$$r_1 = \frac{\sqrt{n\gamma}}{2} \left(1 - \frac{2R \ln 2}{\gamma} - \frac{2 \ln \frac{1-p}{p}}{n\gamma} \right), \quad (14)$$

which represents the normalized distances from the codewords θ and $\mathbf{s}_i$ to their decision boundary. It is important to note that, since $R < C = \frac{1}{2} \log(1 + \gamma) \leq \frac{\gamma}{2 \ln 2}$, we have $1 - \frac{2R \ln 2}{\gamma} > 0$. Therefore, r_1 is positive for sufficiently large n .

For analytical convenience, we categorize the decoding error ε differently from the previous subsection. Let ε_1 denote the error if $\mathbf{x}, \hat{\mathbf{x}} \in \mathcal{C}_n$, where $\hat{\mathbf{x}}$ represents the decoding result. Conversely, let ε_2 denote the error when $\mathbf{x} = \theta$ or $\hat{\mathbf{x}} = \theta$. In other words, ε_1 represents decoding errors within the codebook $\mathcal{C}_n$, while ε_2 represents errors in distinguishing between triggered and idle states. According to the law of total probability, we have

$$P[\varepsilon] = pP[\varepsilon | \mathcal{D}] + (1-p)P[\varepsilon | \bar{\mathcal{D}}] \quad (15)$$

$$= p(P[\varepsilon_1 | \mathcal{D}] + P[\varepsilon_2 | \mathcal{D}]) + (1-p)P[\varepsilon | \bar{\mathcal{D}}]. \quad (16)$$

Shannon derived an approximation for the error rate in finite blocklength coding in [28]. This result was later extended to more general channel models with a rigorous formulation by Polyanskiy in [29]. The key conclusion is summarized in the following lemma, which will be utilized in our subsequent analysis.

Lemma 1: For conventional communication over an AWGN channel with SNR γ and equal-power constraint, the asymptotic expression of the block error rate is given by

$$P_{\text{conv}}[\varepsilon] = Q \left(\sqrt{n} \sqrt{\frac{2(\gamma+1)^2}{\gamma(\gamma+2)}} (C - R) \ln 2 \right), \quad (17)$$

where Q represents the complementary Gaussian cumulative distribution function.

We now proceed to analyze the upper bound of the error rate using Eq. (15). Specifically, Lemma 2 and Lemma 3 establish

upper bounds for the error rate conditioned on the triggered state and idle state, respectively. Building on these results, Theorem 2 provides the upper bound for the overall average error rate $P[\varepsilon]$.

Lemma 2: For ETC over an AWGN channel with SNR γ and equal-power constraint, the block error rate conditioned on the triggered state is bounded by

$$P[\varepsilon | \mathcal{D}] \leq P_{\text{conv}}[\varepsilon] + Q(r_1). \quad (18)$$

Proof: According to the two types of errors previously defined, we first divide the conditional error probability in the triggered state as

$$P[\varepsilon | \mathcal{D}] = P[\varepsilon_1 | \mathcal{D}] + P[\varepsilon_2 | \mathcal{D}]. \quad (19)$$

Now, consider the extended codebook $\mathcal{C}'_n$, which includes an additional codeword θ compared to the conventional codebook $\mathcal{C}_n$. For any transmitted codeword $\mathbf{s}_i$, the decision region corresponding to the first type of error ε_1 becomes smaller compared to conventional communication. Consequently, the average probability of ε_1 is also smaller, i.e.,

$$P[\varepsilon_1 | \mathcal{D}] < P_{\text{conv}}[\varepsilon]. \quad (20)$$

We now turn our attention to ε_2 . Without loss of generality, we assume in the proof that the noise variance is normalized, i.e., $N = 1$ and $P = \gamma$. In the case of equal-power constraint, all codewords are on the surface of an n -dimensional sphere with radius $\sqrt{n\gamma}$. As shown in Fig. 2b, the range of the integral of $P[\varepsilon_2 | \mathcal{D}, \mathbf{s}_i]$ is smaller than the entire half space. Thus, we have

$$P[\varepsilon_2 | \mathcal{D}, \mathbf{s}_i] \leq Q(r_1). \quad (21)$$

The average conditional error rate is bounded by

$$P[\varepsilon_2 | \mathcal{D}] = \sum_{i=1}^M \frac{1}{M} P[\varepsilon_2 | \mathcal{D}, \mathbf{s}_i] \leq Q(r_1) \quad (22)$$

Finally, by substituting Eq. (17) and Eq. (22) into Eq. (19), we arrive at the desired result. ■

Lemma 3: For ETC over an AWGN channel with SNR γ and equal-power constraint, the block error rate conditioned on the idle state has the following upper bounds:

$$P_{\text{ub,high SNR}}[\varepsilon | \bar{\mathcal{D}}] = \frac{\Gamma\left(\frac{n}{2}, \frac{r_0^2}{2}\right)}{\Gamma\left(\frac{n}{2}\right)}, \quad (23)$$

where $\Gamma(\cdot, \cdot)$ denotes the upper incomplete gamma function and $\Gamma(\cdot)$ denotes the gamma function. This bound is more effective in high SNR scenarios. The other upper bound is

$$P_{\text{ub,low SNR}}[\varepsilon | \bar{\mathcal{D}}] = 2^{nR} Q(r_0), \quad (24)$$

which is more effective in low SNR scenarios.

Proof: In the case of an idle transmitter, the upper bound of the error rate can be calculated by scaling down the decision region of θ to the sphere inscribed in the original region, as shown in Fig. 2c. Therefore,

$$P[\varepsilon | \bar{\mathcal{D}}] \leq \Pr\{\chi_n^2 \leq r_0^2\} = \frac{\Gamma\left(\frac{n}{2}, \frac{r_0^2}{2}\right)}{\Gamma\left(\frac{n}{2}\right)}, \quad (25)$$

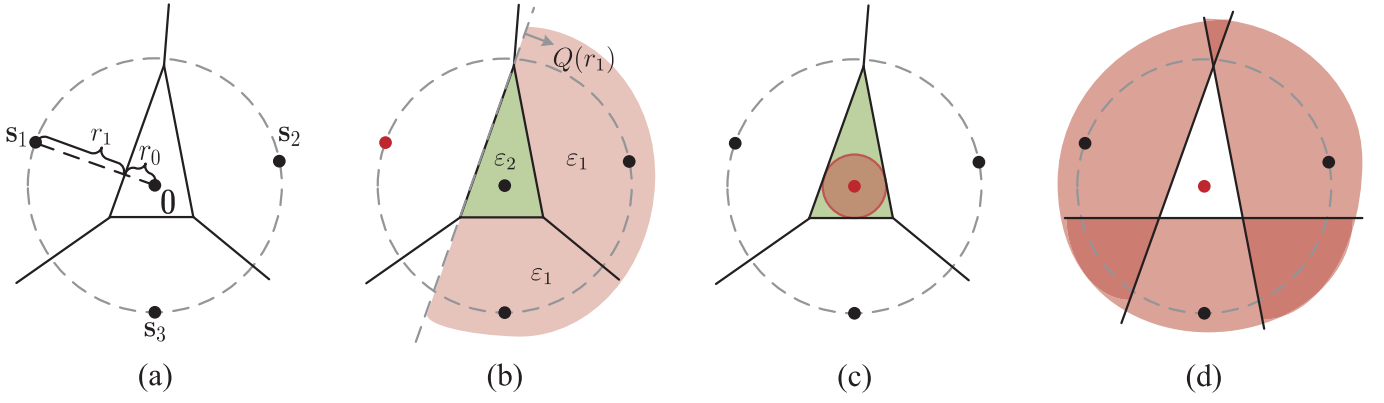


Fig. 2. Schematic of the upper and lower bounds of the error rate.

This upper bound works better under higher SNR, because with a larger codebook size M , the volume of the decision region enclosed by the decision boundaries is closer to the volume of its inscribed sphere.

An alternative upper bound on the error rate can be obtained by summing up the probabilities associated with all half-spaces outside of each decision hyperplane (even if some regions are computed repeatedly), as shown in Fig. 2d. We have

$$P[\varepsilon | \bar{\mathcal{D}}] \leq MQ(r_0) = 2^{nR}Q(r_0). \quad (26)$$

This bound works better under lower SNR conditions because fewer areas are repeatedly calculated at low SNR. ■

Finally, based on Lemma 2, Lemma 3, and Eq. (15), we obtain the following result for the average error rate $P[\varepsilon]$.

Theorem 2: For ETC over an AWGN channel with SNR γ and equal-power constraint, the average error rate is upper bounded by

$$P[\varepsilon] \leq pP_{\text{conv}}[\varepsilon] + pQ(r_1) + (1-p)P_{\text{ub},i \text{ SNR}}, \quad (27)$$

where $i \in \{\text{high}, \text{low}\}$.

Proof: By substituting Eq. (18) and Eq. (23) into Eq. (15), we obtain the results for high SNR scenarios. Similarly, by substituting Eq. (18) and Eq. (24), we obtain the results for low SNR scenarios. ■

Having established the upper bound of the error rate under varying SNR conditions, we now turn our attention to deriving the corresponding lower bound. While the upper bound provides a worst-case estimate of the error probability, the lower bound offers insight into the achievable performance under optimal conditions. By comparing these bounds, we can better understand the range of error rates in the finite blocklength regime and evaluate the efficiency of the proposed ETC scheme.

Theorem 3: For ETC over an AWGN channel with SNR γ and equal-power constraint, the average error rate is bound by

$$P[\varepsilon] \geq pP_{\text{conv}}[\varepsilon] + (1-p)Q(r_0). \quad (28)$$

Proof: Consider the terms in Eq. (15). For $P[\varepsilon | \mathcal{D}]$ in the first term, since ETC involves adding codeword θ to the conventional codebook $\mathcal{C}_n$, the probability of decoding error under the triggered transmitter condition is greater than the

conventional case, i.e.,

$$P[\varepsilon | \mathcal{D}] > P_{\text{conv}}[\varepsilon]. \quad (29)$$

For $P[\varepsilon | \bar{\mathcal{D}}]$ in the second term of Eq. (15), the conditional error rate lower bound is achieved when the decision region of the codeword θ is maximized. This happens when all the codewords $\mathbf{s}_i$ ($i = 1, 2, \dots, M$) coincide at the same position. In this case, the error region of the received symbol is the entire half-space formed by the boundary at a distance r_0 from the origin $\mathbf{0}$. Therefore, the conditional error rate is bounded by

$$P[\varepsilon | \bar{\mathcal{D}}] \geq Q(r_0). \quad (30)$$

Finally, by substituting Eq. (29) and Eq. (30) into Eq. (15), we arrive at the desired result. ■

In this subsection, we derived both upper and lower bounds on the average error rate for a given coding rate R , providing a comprehensive characterization of the error performance in the finite blocklength regime. These bounds are crucial for understanding the performance degradation introduced by ETC in the finite blocklength region. Moreover, they enable the determination of the maximum achievable coding rate for a specified error rate. The numerical results, which further validate and illustrate these theoretical findings, are presented in Section V.

C. Extensions to Fading Channels and Beyond

In this subsection, we extend the channel capacity results for ETC to a broader class of channel models. The capacity proofs for these models generally use Shannon's formula directly and are therefore orthogonal to Theorem 1, making the extension to ETC scenarios straightforward. Below, we provide specific results and explanations. For instance, the capacity results for fading channels from [35] can be readily adapted to ETC using our framework.

Corollary 1: For ETC over fading channels with side information at both the transmitter and receiver, the capacity is achieved through optimal power allocation (i.e., water-filling in time) and is given by

$$C_{\text{ETC}|\mathcal{D}} = \int_{\gamma_0}^{\infty} W \log \left(\frac{\gamma}{\gamma_0} \right) p(\gamma) d\gamma, \quad (31)$$

where $p(\gamma) = \Pr[\gamma[i] = \gamma]$ is the probability distribution of the received SNR, and the parameter γ_0 satisfies

$$\int_{\gamma_0}^{\infty} \left(\frac{1}{\gamma_0} - \frac{1}{\gamma} \right) p(\gamma) d\gamma = 1. \quad (32)$$

For fading channels with side information available at the receiver only, the channel capacity with *i.i.d.* fading is given by

$$C_{\text{ETC}|\mathcal{D}} = \int W \log(1 + \gamma) p(\gamma) d\gamma. \quad (33)$$

For the suboptimal channel inversion scheme, the channel capacity is given by

$$C_{\text{ETC}|\mathcal{D}} = W \log \left(1 + \frac{1}{\mathbb{E}[1/\gamma]} \right). \quad (34)$$

Proof: To establish the capacity for fading channels, it suffices to show that the channel capacity under any given channel gain can be calculated using Shannon's formula. The fading channel model $y = hx + z$ can be reformulated as an AWGN channel with an equivalent noise variance of $N/|h|^2$ by normalizing both sides by the channel gain h . Thus, by Theorem 1, the channel capacity under the triggered condition aligns with Shannon's capacity formula. This completes the proof. ■

Building on the fading channel results, we can further extend the ETC capacity analysis to multiple-input and multiple-output (MIMO) channels. The discrete-time model is given by $\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{z}$, where $\mathbf{x}$ and $\mathbf{y}$ are the transmitted and received symbols of dimensions M_t and M_r , respectively, $\mathbf{z}$ is the M_r -dimensional additive noise, and $\mathbf{H}$ is the $M_r \times M_t$ matrix of channel gains.

Corollary 2: For ETC over MIMO channels, the capacity is given by

$$C_{\text{ETC}|\mathcal{D}} = \max_{\mathbf{R}_\mathbf{x}: \text{Tr}(\mathbf{R}_\mathbf{x})=P} W \log \det [\mathbf{I}_{M_r} + \mathbf{H}\mathbf{R}_\mathbf{x}\mathbf{H}^H], \quad (35)$$

where $\det[\cdot]$ denotes the determinant of the matrix.

The channel capacity results can also be extended to orthogonal frequency-division multiplexing (OFDM); however, the detailed derivation is omitted for brevity. Additionally, according to Theorem 3 in [36], the capacity of an AWGN noncoherent channel, modeled as $y = e^{j\theta}x + z$, approaches the capacity of a coherent channel as the blocklength $N \rightarrow \infty$. This result is equally applicable to ETC because our framework distinguishes between triggered and idle states based solely on signal amplitude, rendering it insensitive to phase variations.

IV. DISTORTION-POWER TRADEOFF OF ETC

In this section, we analyze the average power P_{ave} and average distortion J under the ETC scheme, using the semi-Markov process presented in Section II as an example. We explore two strategies for multi-user channel sharing: statistical multiplexing and joint coding. In statistical multiplexing, multiple users generate data packets independently. In joint coding, a fusion center generates data packets based

on the samples from multiple users. We present the distortion-power tradeoff for both strategies. Additionally, we analyze the optimal number of users to share a channel.

Let K denote the number of independent users sharing the channel. The bandwidth of the common channel is KW , and the total average power is constrained by KP_{ave} . The AWGN channel capacity is given by $C = KW \log \left(1 + \frac{P}{N_0KW} \right)$, where P represents the transmit power and $\frac{N_0}{2}$ represents the power spectral density of the noise. In this section, we assume that the receiver can distinguish whether the transmitter is in a triggered or idle state, so the transmission rate in the triggered state is given by $R_b = C$.

A. Statistical-Multiplexing-Based Multi-User ETC

In this subsection, we focus on the statistical-multiplexing-based multi-user ETC. We start by defining the packet structure. Each packet consists of two parts: address information and state information. The address information, which serves as a packet header identifying the source of each packet, is encoded using $f(K) = \log K$. The state information, representing one of the possible states in S , is encoded using $H(X)$ bits. Thus, the total packet size is $g(K) = H(X) + f(K)$.

Packets are scheduled in a first-in, first-out manner. As the sampling period T_s approaches 0, a new packet is generated and immediately added to the common queue whenever any user changes its state. The packet arrival rate is $\lambda = K/\mathbb{E}[\tau]$. The service rate of the packet is $\mu = KR_b/g(k)$. We define the load factor ρ as

$$\rho = \frac{\lambda}{\mu} = \frac{g(K)}{R_b \mathbb{E}[\tau]}. \quad (36)$$

We model the system as a G/D/1 queue, with the queue length as the state. Let π_i denote the stationary distribution for state i . The following lemmas provide the average power and average distortion of the system.

Lemma 4: For statistical-multiplexing-based multi-user ETC, the average power per user is given by

$$P_{\text{ave}} = (1 - \pi_0)N_0W \left(2^{R_b/W} - 1 \right). \quad (37)$$

Proof: For statistical multiplexing, the probability of the transmitter being triggered is determined by the probability that the queue is not empty, i.e., $p = 1 - \pi_0$. Since the transmit power in the triggered state is obtained by Shannon's formula as $P = N_0W (2^{R_b/W} - 1)$, the average power is given by $P_{\text{ave}} = p \cdot P$. ■

Lemma 5: For statistical-multiplexing-based multi-user ETC, the average distortion is given by

$$J = \left[1 - \sum_{i=0}^{\infty} \left(\frac{K-1}{K} \right)^i \pi_i \right] \Theta. \quad (38)$$

Proof: According to Proposition 1, the average distortion is proportional to the misalignment ratio η . The calculation of η proceeds in two steps. First, we calculate the average misalignment ratio conditioned on the queue length i , denoted by η_i . Next, we average this over different queue lengths.

Consider the case where the queue length is i . Each packet in the queue may originate from one of the K users. Since the

sources of users are *i.i.d.*, the probability that a packet comes from a specific user is $1/K$. Consequently, the probability of no packet originating from a certain user is $(1 - \frac{1}{K})^i$, which also represents the synchronization probability between the transmitter and receiver. The conditional misalignment ratio is given by

$$\eta_i = 1 - \left(\frac{K-1}{K}\right)^i. \quad (39)$$

By substituting Eq. (39) into $\eta = \sum_i \eta_i \pi_i$ and using Eq. (9), we obtain the desired result. ■

Next, we present explicit expressions for the average power and average distortion in the special case of a Poisson arrival.

Theorem 4: For statistical-multiplexing-based multi-user ETC and Poisson arrivals, the average distortion is given by

$$J = \left(1 - \frac{(1-\rho)e^{-\rho/K}}{Ke^{-\rho/K} - K + 1}\right) \Theta, \quad (40)$$

and the average power per user is given by

$$P_{ave} = \rho N_0 W \left(2^{R_b/W} - 1\right). \quad (41)$$

Proof: For the special case of Poisson arrivals, the system becomes an M/D/1 queue. The probability generating function of the stationary distribution is given in [37] by the Pollaczek-Khinchine transform equation as

$$\pi(z) = \frac{(1-z)(1-\rho)h(\lambda(1-z))}{h(\lambda(1-z)) - z}, \quad (42)$$

where $h(s)$ represents the Laplace transform of the service time probability density function. For our M/D/1 queue, $h(s) = \exp(-s/\mu)$. Since $\pi(z)$ is defined as

$$\pi(z) = \mathbb{E}[z^i] = \sum_{i=0}^{\infty} \pi_i z^i, \quad (43)$$

we can take $z = \frac{K-1}{K}$ and obtain

$$\sum_{i=0}^{\infty} \left(\frac{K-1}{K}\right)^i \pi_i = \frac{(1-\rho)e^{-\rho/K}}{Ke^{-\rho/K} - K + 1}. \quad (44)$$

By substituting this result into Eq. (38), we obtain the average distortion J in Eq. (40). By substituting the stationary distribution of the M/D/1 queue $\pi_0 = 1 - \rho$ into Eq. (37), we obtain the average power in Eq. (41). ■

Remark 1: According to the Palm-Khinchine Theorem [38], the superposition of K independent renewal processes becomes asymptotically a Poisson process as K approaches infinity. Therefore, when the number of users is large enough, it is reasonable to use the special case of the M/D/1 queue as an approximation.

Theorem 4 establishes the relationship between the average distortion J and the average power P_{ave} as a parametric equation. Since ρ is a function of R_b , as shown in Eq. (36), J and P_{ave} are both functions of R_b in this parametric equation. For a given average power constraint P_{ave} , the corresponding transmit power P can be determined by solving the following transcendental equation:

$$\lambda g(k) = \frac{P_{ave}}{P} W \log \left(1 + \frac{P}{N_0 W}\right). \quad (45)$$

Then, R_b and J can be calculated based on P .

To this point, we have examined the average distortion and average power for statistical multiplexing-based multi-user ETC. Based on this analysis, a packet-dropping policy can be proposed to further enhance performance. Specifically, whenever a new packet is generated, if the queue contains an undelivered packet from the same user, the stale packet can be discarded. This is because even if the stale packet is transmitted, the estimator will still be misaligned with its source. Implementing this kind of packet-dropping policies reduces both average power consumption and average distortion, thereby enhancing the overall system performance. It is worth noting that if a data packet is being transmitted, then even if the same user changes state again, this data packet cannot be replaced. In other words, the first and last data packets in the system may come from the same user. There are at most $K+1$ data packets in the system at the same time. The following lemma first gives the average distortion of the system under this type of strategy:

Lemma 6: For statistical-multiplexing-based multi-user ETC with the packet-dropping policy and Poisson arrivals, the average distortion is given by

$$J = \frac{1}{K} \left[L - (1 - \pi_0) \left(1 - \frac{1 - e^{-\rho}}{\rho}\right) \right] \Theta, \quad (46)$$

where L denotes the average queue length.

Proof: Similar to the proof of Lemma 5, we proceed in two steps. The difference is that we calculate the conditional average misalignment ratio $\tilde{\eta}_j$ when the packets in the queue come from j different users, rather than when the queue length is i . Consider the scenario where all j packets with different sources originate from the other $K-1$ users, there are $\binom{K-1}{j}$ ways to allocate them because any two packets come from different users. Conversely, if there are no restrictions on the source of the packets, there are $\binom{K}{j}$ ways. Thus, the conditional misalignment ratio can be expressed as

$$\tilde{\eta}_j = 1 - \frac{\binom{K-1}{j}}{\binom{K}{j}} = \frac{j}{K}. \quad (47)$$

It is important to note that j will not exceed K under the packet-dropping policy. Therefore, the average misalignment probability η is given by

$$\eta = \sum_{j=0}^K \tilde{\eta}_j \tilde{\pi}_j = \frac{1}{K} \sum_{j=0}^K j \tilde{\pi}_j, \quad (48)$$

where $\tilde{\pi}_j$ denote the stationary distribution for the number of packets with different sources in the system. Notice that the sum term $\sum_{j=0}^K j \tilde{\pi}_j$ can be calculated by making a correction to the average queue length $L = \sum_{i=0}^K i \pi_i$ as

$$\sum_{j=0}^K j \tilde{\pi}_j = \left(\sum_{i=0}^K i \pi_i \right) - (1 - \pi_0) \tilde{p}, \quad (49)$$

where $\tilde{p}$ the conditional probability that the last packet in the queue and the packet being transmitted come from the same source when the system is in a busy period. Due to the fixed service rate and the memorylessness of the Poisson process,

$\tilde{p}$ can be calculated by integrating a truncated exponential distribution $\text{Exp}(\lambda)$ within a packet service time $1/\mu$ as

$$\tilde{p} = \int_0^{1/\mu} \frac{1-x}{1/\mu} \cdot \lambda \exp(-\lambda x) dx \quad (50)$$

$$= 1 - \frac{1 - e^{-\rho}}{\rho}. \quad (51)$$

Finally, by substituting Eq. (48), Eq. (49), and Eq. (51) into Eq. (9), we obtain the desired result. ■

The above lemma shows how the average distortion and average power of the system depend on the average queue length L and stationary distribution π_i . The following theorem further presents the analytical expressions of average distortion and average power.

Theorem 5: For statistical-multiplexing-based multi-user ETC with the packet-dropping policy and Poisson arrivals, the average distortion is given by

$$J = \frac{1}{K} \left[\left((K+1) - \frac{1}{\rho}(1-\hat{e}) \right) \left(1 + \frac{1}{(K+1)\lambda\mathbb{E}[b] + K} \right) - \frac{\lambda\mathbb{E}[b]}{\lambda\mathbb{E}[b] + 1} \left(1 - \frac{1 - e^{-\rho}}{\rho} \right) \right] \Theta, \quad (52)$$

and the average power per user is given by

$$P_{ave} = \frac{\lambda\mathbb{E}[b]}{\lambda\mathbb{E}[b] + 1} N_0 W \left(2^{R_b/W} - 1 \right). \quad (53)$$

where $\mathbb{E}[b]$ denotes the average length of the busy period b . $\mathbb{E}[b]$, $\phi(m)$, and $\hat{e}$ are given by

$$\mathbb{E}[b] = \frac{1}{\mu} \sum_{i=0}^K \binom{N}{i} \frac{1}{\phi(i)}, \quad (54)$$

$$\phi(m) = \begin{cases} \prod_{i=1}^m \frac{\exp(-i\rho)}{1 - \exp(-i\rho)}, & m \neq 0 \\ 1, & m = 0 \end{cases} \quad (55)$$

$$\hat{e} = \left[1 + K\rho \sum_{i=1}^K \binom{K}{i} \frac{1}{\phi(i)} \right]^{-1}, \quad (56)$$

respectively.

Proof: In this proof, we apply the conclusion of the finite-source queue model to obtain the average queue length of the system, and further substitute the expression into Lemma 6 to complete the proof. Specifically, note that when the queue length $i \neq 0$, the actual arrival rate of the queue is given by $(K+1-i)\lambda_1$, where $\lambda_1 = \lambda/K = 1/\mathbb{E}[\tau]$ denote the arrival rate of a single user. This is because, except for the packet being transmitted, the source users of all packets already in the queue no longer contribute to the arrival rate. Therefore, during the busy period, the system can be modeled as finite-source model with $K+1$ users. On the other hand, the average duration of the idle period is given by $1/\lambda$ according to the property of Poisson arrival.

According to conclusions from Chapter II of [39], the average length of a busy period is given by Eq. (54), and the average queue length of a finite-source model with $K+1$ user is given by

$$\hat{L} = (K+1) - \frac{1}{\rho}(1-\hat{e}), \quad (57)$$

where $\hat{e}$ is given by Eq. (56) and Eq. (55). Note that for the average queue length of $K+1$ users, we use a different symbol $\hat{L}$ because its average length of an idle period is different from our system. With the following correction, the average queue length L is given by

$$L = \hat{L} \cdot \frac{\mathbb{E}[b]}{\mathbb{E}[b] + \frac{1}{(K+1)\lambda_1}} \cdot \frac{\mathbb{E}[b] + \frac{1}{K\lambda_1}}{\mathbb{E}[b]} \quad (58)$$

$$= \hat{L} \left(1 + \frac{1}{(K+1)\lambda\mathbb{E}[b] + K} \right). \quad (59)$$

By substituting Eq. (57) and Eq. (59) into Eq. (46), we obtain the average distortion in Eq. (52).

Based on the results of the busy period average length, the stationary probability for an empty queue is given by

$$\pi_0 = \frac{1/\lambda}{\mathbb{E}[b] + 1/\lambda} = \frac{1}{\lambda\mathbb{E}[b] + 1}. \quad (60)$$

By substituting Eq. (60) into Eq. (37), we obtain the average power in Eq. (53). ■

Remark 2: As mentioned in Section I, The ETC we are concerned with is a cross-layer strategy involving the physical layer, and can be used simultaneously with the existing AoII-aware sampling strategy. Actually, the conclusions of Theorem 5 also applies to the AoII-aware sampling strategy, that is, whenever the queue is emptied, the user with the largest non-zero AoII is selected for sampling, encoding and a packet transmission. The reason why AoII-aware sampling and the packet-dropping policy share the same average distortion and average power is that the states of the semi-Markov process are randomly taken from the fixed set S . The two strategies will have different performance if we consider stochastic processes with independent increments such as the Wiener process, but this is beyond the scope of this paper.

B. Joint-Coding-Based Multi-User ETC

In this subsection, we focus on joint-coding-based multi-user ETC, where the states of all users are encoded into a single data packet at the fusion center. Transmission decisions are made periodically. Specifically, the current samples are compared with the samples from the last period, as shown in Eq. (11). Packets are generated only when any user's state changes during the last period. Unlike the previous subsection, the sampling period T_s cannot approach 0, as this would be equivalent to independently generating data packets. Joint coding is expected to reduce the amount of transmitted information because address information can be compressed when multiple users change their states in the same period. Define p_s as the probability that a user changes its state between two consecutive samples. The following lemma provides the expression for the average power P_{ave} .

Lemma 7: For joint-coding-based multi-user ETC, the average power per user is given by

$$P_{ave} = N_0 W \left(1 - (1 - p_s)^K \right) \left(2^{\frac{q}{KWT_s}} - 1 \right), \quad (61)$$

where $g = g(K, p_s)$ represents the average packet size of the joint coding and is given by

$$g = - \sum_{i=1}^K \binom{K}{i} \frac{p_s^i (1-p_s)^{K-i}}{1 - (1-p_s)^K} \log \frac{p_s^i (1-p_s)^{K-i}}{(1 - (1-p_s)^K) H^i(X)} \quad (62)$$

Proof: To analyze the average power of joint coding, we first calculate the entropy of the sample conditioned on at least one active user, thereby determining the theoretical size of a packet. The number of users that change their states during a sampling period T_s can be represented using the binomial distribution:

$$\Pr\{v = i \mid v \neq 0\} = \binom{K}{i} \frac{p_s^i (1-p_s)^{K-i}}{1 - (1-p_s)^K}, \quad (63)$$

where v denotes the number of active users. The size of the jointly encoded packet is given by

$$g(K) = \sum_{i=1}^K -\Pr\{v = i \mid v \neq 0\} \log \frac{\Pr\{v = i \mid v \neq 0\}}{\binom{K}{i} H^i(X)}. \quad (64)$$

Substituting Eq. (63) into Eq. (64) yields Eq. (62). Finally, since the transmission rate per user satisfies $R_b = \frac{g}{kT_s}$, the average power consumption is given by

$$P_{ave} = \Pr\{v \neq 0\} \cdot N_0 W \left(2^{\frac{g}{kT_s}} - 1 \right). \quad (65)$$

Substituting the probability of an active transmitter $\Pr\{v \neq 0\} = 1 - (1-p_s)^K$ yields Eq. (61). ■

Remark 3: The expression for the packet size $g(K)$ highlights how joint coding efficiently compresses address information. For example, in statistical multiplexing, if two users are active within a period, the address information size is $2 \log k = \log k^2$. In contrast, for joint coding, the joint address information size is $\log \binom{k}{2} = \log \frac{k^2 - k}{2}$. This reduction in size stems from eliminating the order information of multiple users.

Lemma 8: For joint-coding-based multi-user ETC, if the sampling rate is much greater than the arrival rate λ , i.e., $T_s \ll \mathbb{E}[\tau]$, the average distortion can be approximated by

$$J \approx \frac{3}{2} p_s \Theta. \quad (66)$$

Proof: When $T_s \ll \mathbb{E}[\tau]$ (i.e., $p_s \ll 1$), it is highly unlikely for two transitions to occur within the same period. Under these conditions, the conditional distribution of each transition can be approximated by a uniform distribution over the period. Consequently, each transition introduces an average misalignment time of $\frac{1}{2} T_s$. Additionally, we account for the transmission delay of T_s in ETC. Therefore, the average misalignment ratio is given by

$$\eta \approx p_s \cdot \left(\frac{1}{2} + 1 \right) T_s / T_s = \frac{3}{2} p_s, \quad (67)$$

which leads to the result in Eq. (66). ■

Theorem 6: For joint-coding-based multi-user ETC and Poisson arrivals, the average distortion is given by

$$J = \left(1 - \frac{e^{-\lambda_s} (1 - e^{-\lambda_s})}{\lambda_s} \right) \Theta, \quad (68)$$

and the average power per user is obtained by substituting $p_s = 1 - e^{-\lambda_s}$ into Eq. (61).

Proof: To analyze the average distortion, we examine when the estimator aligns with the source updates. It should be noted that due to the consideration of transmission delay, the data packets encoded at the end of the period will be transmitted in the next period. This alignment condition can be divided into two parts: First, the user is not active in the last period, otherwise a new state is still being transmitted during the current period. Second, the user's source has not changed state in the current period. The calculation proceeds in two steps.

Let u denote the number of times the user's state changes in a period. The probability of the first condition is given by $\Pr\{u = 0\}$. For the second condition, we focus on the distribution of state transitions within a period. Given the number of arrivals $u = i$ in a period, the conditional distribution of these i events is equivalent to the order statistics of i i.i.d. uniform random variables. The misalignment ratio η is only related to the distribution of the first transition in a period, the cumulative distribution function of which is given by

$$F_{(1)}(t) = 1 - [1 - F(x)]^i = 1 - \left(1 - \frac{t}{T_s} \right)^i. \quad (69)$$

Therefore, the conditional misalignment ratio is given by

$$\eta_{u=i} = \frac{1}{T_s} \int_0^{T_s} F_{(1)}(t) dt = \frac{i}{i+1} \quad (70)$$

for $i \neq 0$. Additionally, we have $\eta_{u=0} = 0$. Therefore, by substituting the probability function of the Poisson distribution, we have

$$\eta = 1 - \Pr\{u = 0\} \sum_{i=0}^{\infty} (1 - \eta_{u=i}) \Pr\{u = i\} \quad (71)$$

$$= 1 - e^{-2\lambda_s} \sum_{i=0}^{\infty} \frac{\lambda_s^i}{(i+1)!} \quad (72)$$

$$= 1 - \frac{e^{-\lambda_s} (1 - e^{-\lambda_s})}{\lambda_s}. \quad (73)$$

By substituting η into Eq. (9), we obtain Eq. (68). To calculate the average power P_{ave} , the parameter p_s is given by $p_s = 1 - \Pr\{u = 0\} = 1 - e^{-\lambda_s}$. This completes the proof. ■

Theorem 6 establishes the relationship between J and P_{ave} as a parametric equation. Since p_s in the expression of P_{ave} in Eq. (61) is determined by λ_s , both J and P_{ave} are both functions of λ_s in this parametric equation. Furthermore, in the joint coding scenario, TDMA can be employed to minimize transmission delay by utilizing the channel periodically. We assume that data packets generated by each sample can be transmitted immediately with zero delay under TDMA. This leads to the following theorem.

Theorem 7: For joint-coding-based multi-user ETC with TDMA and Poisson arrivals, the average distortion is given by

$$J = \left(1 - \frac{1 - e^{-\lambda_s}}{\lambda_s} \right) \Theta. \quad (74)$$

Proof: This proof follows a similar approach to Theorem 6 but is more straightforward. With the assumption of zero transmission delay, the misalignment ratio η depends solely on u of the current period. By applying the Poisson distribution, we have

$$\eta = \sum_{i=0}^{\infty} \eta_{u=i} \Pr\{u = i\} \quad (75)$$

$$= e^{-\lambda_s} \sum_{i=1}^{\infty} \frac{\lambda_s^i}{(i-1)!(i+1)} \quad (76)$$

$$= 1 - \frac{1 - e^{-\lambda_s}}{\lambda_s}. \quad (77)$$

Substituting η into Eq. (9) yields the result. ■

It is important to note that we do not provide a specific codebook and coding scheme for the joint coding method. Instead, we utilize information theory to estimate the packet size. As a result, the derived performance metrics serve as upper bounds for the joint coding method.

C. Optimal Number of Users

In this subsection, we investigate the optimal number of users, denoted as K^* , in the multi-user scenario where the maximum transmission rate is fixed at R_b . It is important to note that increasing the number of users does not always lead to better system performance due to the overhead introduced by address information. Using statistical multiplexing-based multi-user ETC as an example, we examine the properties of the optimal number of users in two extreme cases: $K^* \rightarrow \infty$ and $K^* = 1$. First, we analyze the growth rate of K^* relative to the compression ratio of the address information $f(K)$. Second, we provide necessary conditions under which $K^* = 1$. As a preliminary step, the following lemma presents an asymptotic upper bound for the average distortion J .

Lemma 9: For statistical-multiplexing-based multi-user ETC and Poisson arrivals, the average distortion is bounded by

$$J < \frac{1}{K} \left(\rho + \frac{\rho^2}{2(1-\rho)} \right) \Theta. \quad (78)$$

This upper bound also approximates the average distortion J . As the number of users K approaches infinity, the gap between J and its upper bound asymptotically approaches 0.

Proof: Using Eq. (39), and noting that $(1-x)^i > 1-ix$ for $0 < x < 1$, we have

$$\eta_i = 1 - \left(1 - \frac{1}{K} \right)^i < \frac{i}{K}. \quad (79)$$

Therefore,

$$\eta < \frac{1}{K} \sum_{i=1}^{\infty} i \pi_i = \frac{L}{K} = \frac{1}{K} \left(\rho + \frac{\rho^2}{2(1-\rho)} \right), \quad (80)$$

where L denotes the average number of packets in the system. Substituting η into Eq. (9) yields the result in Eq. (78). Next, we analyze the scenario when the number of users tends to infinity. Taking the limit as $K \rightarrow \infty$ in Eq. (38), we obtain

$$\lim_{K \rightarrow \infty} J = \lim_{K \rightarrow \infty} \Theta \sum_{i=0}^{\infty} \left(1 - \left(\frac{K-1}{K} \right)^i \right) \pi_i = \frac{L}{K} \Theta. \quad (81)$$

This expression matches the upper bound found previously, indicating that the upper bound serves as an accurate approximation of the average distortion J when K is large. The approximation error tends to zero as K increases. This completes the proof. ■

Based on the above approximation of J , we can intuitively understand the relationship between the average distortion, the number of users, and the address information. Recalling the expression for the system load factor given in Eq. (36), we observe that ρ increases monotonically with K and may exceed 1. When $\rho \geq 1$, the queue diverges, leading to an infinite average queue length L and a maximum average distortion $J = \Theta$. Conversely, if the receiver can naturally identify the source of packets, allowing the omission of address information (i.e., $f(K) = 0$), the performance improves monotonically with increasing K .

In practical systems, diversity gain and timing information facilitate the compression of address information. For instance, if the holding time of the semi-Markov process is a fixed constant, we can accurately determine the source of a newly arrived packet based on the arrival time of historical packets. In this subsection, we do not delve into specific address information compression schemes. Instead, we focus on the relationship between the optimal number of users K^* and the compression ratio β , defined as

$$\beta = \frac{\log K}{f(K)}. \quad (82)$$

This implies that the size of address information is compressed to $1/\beta$ of its original size.

Theorem 8: For statistical-multiplexing-based multi-user ETC with a fixed transmission rate and Poisson arrivals, the optimal number of users increases exponentially with the compression ratio β and can be approximated as

$$K^* \approx \exp [\beta \ln 2 (R_b \mathbb{E}[\tau] - H(X)) - 1] \quad (83)$$

for large β . The error of the approximation asymptotically approaches 0 as β approaches infinity.

Proof: First, we note that $K^* \rightarrow \infty$ as $\beta \rightarrow \infty$. Although the number of users can only be integers, we are concerned with the growth rate as K^* tends to infinity. Therefore, we treat K as a continuous variable. To minimize distortion, take the derivative of the misalignment ratio upper bound η_{ub} in Eq. (80) with respect to K to obtain

$$\frac{d\eta_{ub}}{dK} = \frac{K\rho'(\rho^2 - 2\rho + 2) - \rho(\rho - 1)(\rho - 2)}{2K^2(1 - \rho)^2}, \quad (84)$$

where $\rho' = \frac{d\rho}{dK}$. Setting the derivative to 0, we have

$$K\rho' = \frac{\rho(\rho - 1)(\rho - 2)}{\rho^2 - 2\rho + 2}. \quad (85)$$

Substituting the expression for ρ , we get

$$K\rho' = \frac{Kf'(K)}{R_b \mathbb{E}[\tau]} = \frac{1}{\beta \ln 2} \cdot \frac{1}{R_b \mathbb{E}[\tau]}. \quad (86)$$

As $\beta \rightarrow \infty$, the right side of Eq. (86) tends to 0. Comparing Eq. (85) and Eq. (86), and noting that $\rho^2 - 2\rho + 2 > 1$ and $\rho - 2 < -1$ in Eq. (85), we find that $\rho(\rho - 1) \rightarrow 0$ as $\beta \rightarrow \infty$.

Since $\lim_{\beta \rightarrow \infty} \rho = 0$ is impossible, we have $\lim_{\beta \rightarrow \infty} \rho = 1$. Therefore, for sufficiently large β ,

$$\rho(K^*) \approx 1 - \frac{1}{\beta \ln 2} \cdot \frac{1}{R_b \mathbb{E}[\tau]}. \quad (87)$$

Finally, substituting $\rho(K^*)$ into Eq. (87) yields Eq. (83), with the error approaching 0 as $\beta \rightarrow \infty$. ■

Corollary 3: When the compression ratio $\beta \rightarrow \infty$, the optimal number of users is approximately $\frac{1}{e} K_{\max}$, where $K_{\max}$ represents the maximum number of users to maintain queue convergence.

Proof: When the number of users reaches the maximum while maintaining queue convergence, the load factor can be approximated by $\rho(K_{\max}) \approx 1$. Thus,

$$K_{\max} \approx \exp[\beta \ln 2 (R_b \mathbb{E}[\tau] - H(X))] \quad (88)$$

By comparing Eq. (83) and Eq. (88), we can find that $K^* \approx \frac{1}{e} K_{\max}$. ■

Next, let's examine the necessary conditions for $K^* = 1$. In other words, we will analyze the scenario where there is no benefit for multiple users to share a common channel.

Theorem 9: For statistical-multiplexing-based multi-user ETC with a fixed transmission rate and Poisson arrivals, the following necessary condition must hold for the optimal number of users to be $K^* = 1$:

$$R_b \mathbb{E}[\tau] < \frac{[H(X) + 1]^2}{H(X) - 1}. \quad (89)$$

Proof: To establish that $K^* = 1$, the average distortion for $K = 1$ must be less than or equal to the average distortion for $K = 2$. Using the expression for average distortion from Eq. (40), we have

$$\frac{1 - \rho(1)}{1 - \rho(2)} \geq \frac{e^{-\rho(2)/2}}{2e^{-\rho(2)/2} - 1}. \quad (90)$$

Using the series expansion to get $\frac{e^{-x}}{2e^{-x}-1} > 1 + x$ and substituting $\rho(1) = \frac{H(X)}{R_b \mathbb{E}[\tau]}$ and $\rho(2) = \frac{H(X)+1}{R_b \mathbb{E}[\tau]}$, we get

$$1 + \frac{1}{R_b \mathbb{E}[\tau] - H(X) - 1} > 1 + \frac{1}{2} \cdot \frac{H(X) + 1}{R_b \mathbb{E}[\tau]}. \quad (91)$$

By simplifying this inequality, we obtain the desired result. ■

The key parameter, $R_b \mathbb{E}[\tau]$, represents the average number of bits that can be transmitted per trigger. Intuitively, we can infer that a single-user exclusive channel performs better when the transmission capacity is relatively low. Conversely, when communication resources are abundant, a multi-user shared channel offers superior performance.

V. NUMERICAL RESULTS

In this section, we present numerical results to demonstrate the potential of ETC and validate the theoretical analysis from the previous sections. The performance in various scenarios is compared to provide valuable insights. Specifically, we first compare the channel coding rate of ETC and conventional communication in the finite blocklength regime, illustrating the reduction in coding rate achieved by introducing ETC. Next,

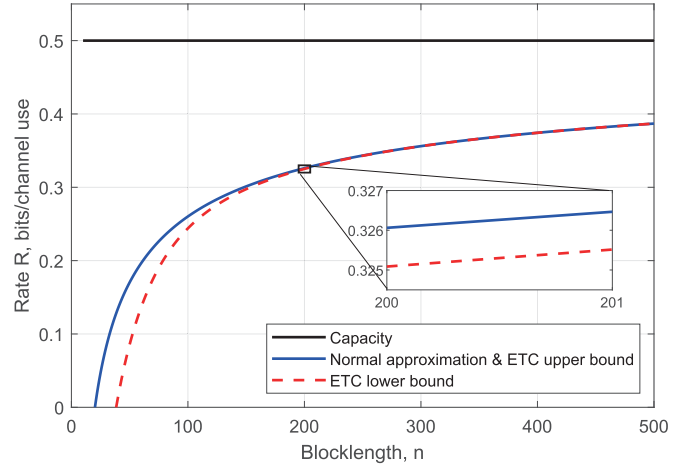


Fig. 3. Channel coding rate vs. blocklength for the AWGN channel and ETC with trigger probability $p = 0.5$, SNR = 0 dB, block error rate $\epsilon = 10^{-3}$.

we compare the performance of ETC with traditional periodic communication for a specific class of semi-Markov processes, revealing the advantages of ETC over periodic communication. Finally, we extend the analysis to the multi-user scenario, comparing the performance of ETC based on statistical multiplexing and joint coding. Additionally, we examine the relationship between average distortion and the number of users.

A. Finite Blocklength ETC

Fig. 3 illustrates the rate-blocklength tradeoff for ETC over an AWGN channel. The trigger probability is set to $p = 0.5$, and the maximal block error rate in the finite blocklength regime is $\epsilon = 10^{-3}$. The black line represents the channel capacity $C = 0.5$. The blue line shows the normal approximation of the AWGN channel coding rate in the finite blocklength regime, obtained from Eq. (296) in [29]. For a fair comparison between ETC and conventional communication, we set the ETC maximal conditional block error rate in both the triggered and idle states to ϵ , equal to the upper bound of $P_{\text{conv}}[\epsilon]$. In other words, we set $P[\epsilon | \mathcal{D}] \leq \epsilon$, $P[\epsilon | \bar{\mathcal{D}}] \leq \epsilon$, and $P_{\text{conv}}[\epsilon] \leq \epsilon$. The red line in the figure represents the performance lower bound of the ETC channel coding rate conditioned on the triggered state.

The results indicate that in the finite blocklength regime, the channel coding rates for both ETC and conventional communication converge to the channel capacity C as the blocklength n tends to infinity. Additionally, the lower bound of the ETC coding rate converges to the result of the normal approximation at an even faster rate as n increases. We can see from the zoom-in sub-figure that when the blocklength $n \approx 200$, the upper and lower bounds of the channel capacity for ETC differ by approximately 3%. Consequently, the channel capacity derived from the normal approximation in traditional finite blocklength coding theory serves as a reliable estimate for the channel capacity of ETC.

In summary, these findings confirm that for sufficiently large blocklengths, ETC can achieve the same conditional

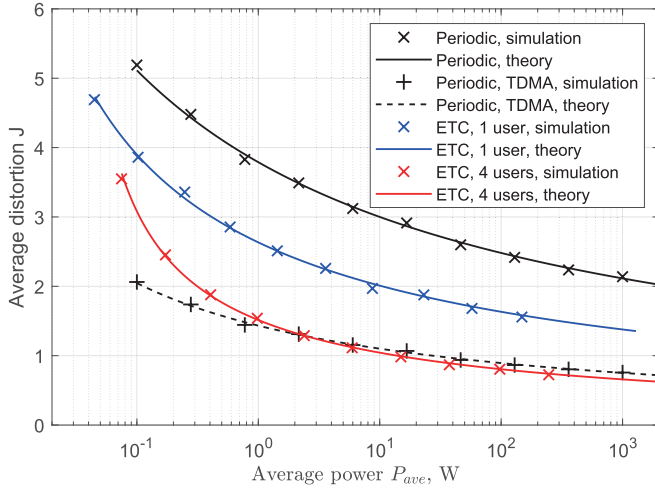


Fig. 4. Average distortion vs. average power per user. Comparison between ETC and periodic communication. $W = 1$ Hz.

coding rate as conventional communication without incurring additional communication costs. In other words, the cost of distinguishing the transmitter state in ETC becomes negligible for large blocklengths.

B. Performance Comparison of Real-Time Monitoring

In this subsection, we evaluate the performance of various real-time monitoring and reconstruction strategies using semi-Markov stochastic processes as an example. The state of the source is chosen from the set $S = \{0, 1, 2, \dots, 7\}$ with equal probability, i.e., $p_i = \frac{1}{8}$ for $i = 1, 2, \dots, 8$. The holding time τ of the process follows an exponential distribution, $\tau \sim \text{Exp}(1)$, meaning that the counting process of state changes is a Poisson process with an arrival rate of $\lambda_1 = 1$. For the AWGN channel, the power spectral density of the additive noise is set to $N_0 = 10^{-3}$.

The performance of ETC and traditional periodic communication under different bandwidths are compared in Fig. 4 and Fig. 5. In this context, ETC refers to statistical-multiplexing-based ETC. The black lines in the figures represent the performance of periodic communication. At the beginning of each period T_s , the sampled source state information is encoded into a packet of $H(X)$ bits and transmitted during this period. For periodic channel use, the Time-Division Multiple Access (TDMA) method can be applied in multi-user scenarios to reduce transmission delays. The dotted lines in the figures represent the performance of periodic transmission with TDMA when transmission delay is ignored, which is a performance upper bound for periodic communication.

The results demonstrate that ETC significantly outperforms traditional periodic communication, especially when the bandwidth is large. As illustrated in Fig. 5, ETC only requires 10% or even 1% of the average power to achieve the same average distortion as periodic communication. When the bandwidth is relatively small ($W = 1$ Hz), ETC still outperforms standard

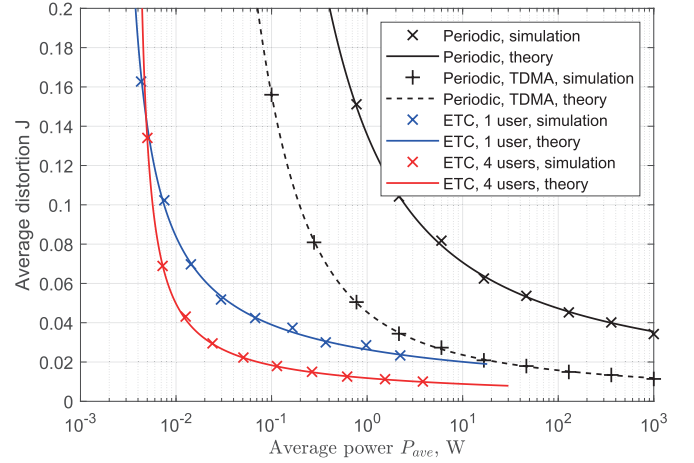


Fig. 5. Average distortion vs. average power per user. Comparison between ETC and periodic communication. $W = 100$ Hz.

periodic communication but does not perform as well as periodic communication using TDMA, as shown in Fig. 4. This result is understandable given that periodic communication with TDMA makes the strong assumption that the transmission delay is equal to 0, a condition achievable only as the number of users approaches infinity, which is typically impractical. Additionally, the performance of a 4-user shared channel ETC surpasses that of the single-user case at both bandwidths, indicating that sharing communication resources among an appropriate number of users enhances performance. Detailed results regarding the optimal number of users are presented in the next subsection.

Fig. 6 illustrates the relationship between average distortion and average power using statistical multiplexing and joint coding, with the number of users set to $K = 4$. The solid lines in the figure represent the theoretical results, and the x-shaped markers represent the simulation results. The yellow and red lines correspond to the statistical-multiplexing-based ETC with and without a packet dropping policy, while the blue and black lines represent the scenario without ETC. In the absence of our ETC methods, a channel with a constant bit rate R_b is established, and the encoding method is structured as follows: when the queue is empty, a “0” is transmitted; when the queue is not empty, the data packet begins with one bit of “1”, followed by $H(X)$ bits of state information and $\log K$ bits of address information. The size of a data packet in the non-ETC scenario is given by $g_{\text{non-ETC}}(K) = H(X) + \log K + 1$.

We first focus on the comparison between the four statistical multiplexing strategies. Both ETC and the packet-dropping policy enhance system performance. Notably, as mentioned in Section IV-A, the strategy incorporating packet dropping is equivalent to AoII-aware sampling in terms of performance. Therefore, Fig. 6 also compares the impacts of ETC and AoII-aware strategies on overall performance. It is evident that performance is maximized when both ETC and AoII-aware sampling are utilized. However, if only one method

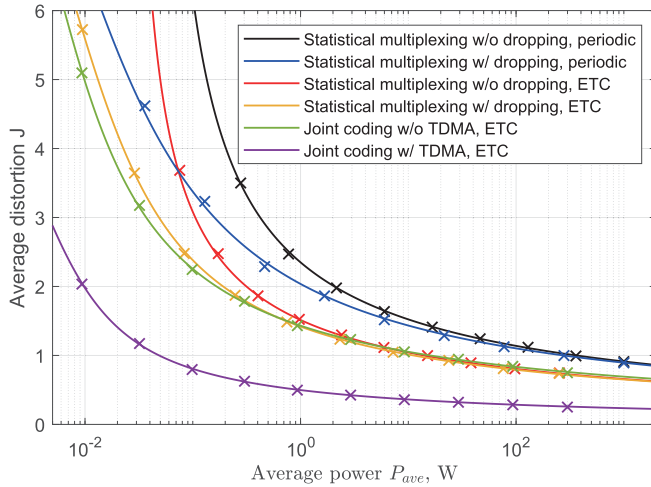


Fig. 6. Average distortion vs. average power per user with statistical-multiplexing based and joint-coding based ETC.

is employed, ETC provides a more significant performance improvement when the average power constraint is large (for example, when $P_{ave} < 7 \times 10^{-2} W$ in the figure), whereas AoI-aware sampling yields greater performance enhancement under relatively small average power constraints.

Next we turn our attention to the comparison between statistical multiplexing and joint coding. It can be observed that joint-coding-based ETC outperforms statistical-multiplexing-based ETC when the average power is low (e.g., when $P_{ave} < 2 \times 10^{-1} W$ in the figure). This result is consistent with our expectations, as the joint coding method benefits from a fusion center and additional state information. Conversely, when the average power constraint is high, there is no significant difference between the performances of statistical multiplexing and joint coding. This suggests that the advantage of joint coding for ETC diminishes when ample communication resources are available. Additionally, the purple line in the figure corresponds to the joint coding method using TDMA, where the transmission delay is neglected. This represents the performance limit when the number of users approaches infinity and can be regarded as a performance upper bound.

C. Optimal Number of Users

In this subsection, we verify the relationship between ETC performance and the number of users K . We set the average power constraint to $P_{ave} = 0.2W$. Fig. 7 illustrates the relationship between average distortion and the number of users at three different bandwidths for statistical-multiplexing-based multi-user ETC. In the case of $W = 0.6$ Hz, the average distortion increases with the number of users, indicating that a single-user exclusive channel offers optimal performance. For $W = 0.8$ Hz and $W = 1$ Hz, the average distortion initially decreases and then increases as the number of users increases. This implies the existence of an optimal number of users, $K^* > 1$, that minimizes the average distortion. This

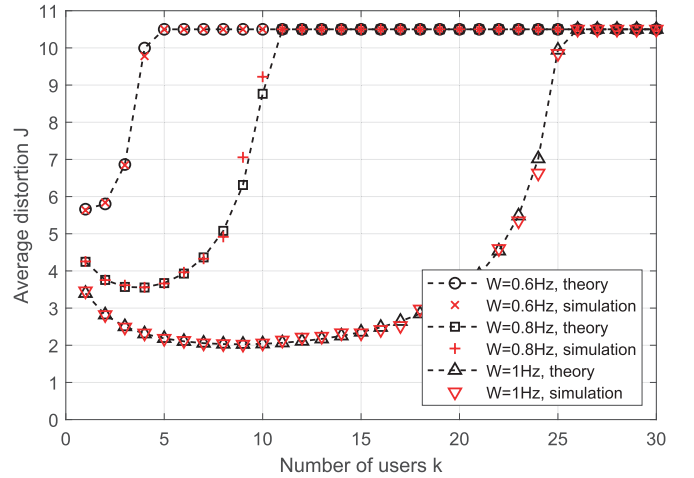


Fig. 7. Average distortion vs. the number of users.

observation is consistent with the theoretical result that a larger $R_b \mathbb{E}[\tau]$ corresponds to a greater optimal number of users K^* .

It should be noted that at points on the curve where distortion rises rapidly with the number of users, the simulation results slightly deviate from the theoretical predictions. This discrepancy occurs because the load factor ρ at these points is very close to 1, causing the queue to approach divergence. Consequently, the average queue length becomes large, and the numerical results are unstable.

VI. CONCLUSION

In this paper, we explored the feasibility and performance of event-triggered communication for the real-time reconstruction of stochastic processes. By equating the idle state to an all-zero codeword and utilizing an extended codebook, we analyzed the channel coding rate in both infinite and finite blocklength regimes. Notably, we demonstrated that even with the additional codeword in ETC, the rate of infinite blocklength coding remains unchanged, and Shannon's formula is still applicable. In the finite blocklength regime, we provided performance upper and lower bounds for ETC. These results are then further extended to more general models such as fading channels.

We then used the real-time reconstruction of a semi-Markov process as a case study to evaluate the performance of ETC. Our estimation strategy showed that minimizing the average distortion is equivalent to minimizing the misalignment time ratio. We further examined ETC in a multi-user scenario, considering both statistical multiplexing and joint coding approaches. Specifically, we derived explicit distortion-power tradeoffs in the form of parametric equations for Poisson arrivals. Finally, our numerical results validated the theoretical findings, demonstrating that ETC causes negligible channel capacity degradation for larger blocklengths and significantly outperforms traditional periodic communication.

Future research in ETC may focus on its applications in smart grid and multi-agent systems, expanding its potential utility in various real-world scenarios.

APPENDIX
 PROOF OF THEOREM 1

In this proof, we construct random codebooks for an ensemble of systems, each consisting of a transmitter, channel, and receiver. We introduce a two-step decoding scheme and analyze the average error rate $P[\varepsilon]$. We prove the theorem by showing that $P[\varepsilon] < \epsilon$ for sufficiently large n . Without loss of generality, we assume that the noise variance is normalized, i.e., $N = 1$ and $P = \gamma$. Taking the maximal power constraint as an example, all the codewords are confined within an n -dimensional sphere with radius $\sqrt{nP}$. To maintain the sphere size independent of n , we further normalize the vectors to

$$\underline{s}_i \triangleq \mathbf{s}_i / \sqrt{n}; \quad i = 1, 2, \dots, M, \quad (92)$$

$$\underline{z} \triangleq \mathbf{z} / \sqrt{n}, \quad (93)$$

$$\underline{y} \triangleq \mathbf{y} / \sqrt{n} = \underline{s}_i + \underline{z}. \quad (94)$$

We construct our random codebooks using a method similar to [40]. The codes of an ensemble of systems can be specified by a probability density function of codewords $\underline{s}_i$. A simple choice that satisfies the maximal power constraint is given by

$$p_{\underline{s}_i}(\alpha) = \begin{cases} \frac{1}{V_s}, & |\alpha|^2 \leq P \\ 0, & |\alpha|^2 > P \end{cases}, \quad \forall i, \quad (95)$$

where V_s denotes the volume of the signal sphere. Consequently, the probability of a source signal falling within a specific region is proportional to the volume of that region. For any $\Delta > 0$,

$$\Pr \left\{ |\underline{s}_i| \leq \sqrt{P - \Delta} \mid \mathcal{D} \right\} \quad (96)$$

$$= \frac{\text{volume of sphere of radius } \sqrt{P - \Delta}}{\text{volume of sphere of radius } \sqrt{P}} \quad (97)$$

$$= \frac{(\sqrt{P - \Delta})^n}{(\sqrt{P})^n} = \left(1 - \frac{\Delta}{P}\right)^{n/2}, \quad (98)$$

which approaches 0 as n becomes large. It is consistent with the fact that almost all of the volume in a high-dimensional sphere is located near the surface.

The decoding process involves first determining whether the transmitter is in a triggered state by comparing the amplitude of the received symbol $\mathbf{y}$ with a given threshold $y_{th} = \sqrt{n \left(\frac{P}{2} + 1\right)}$. If the transmitter is identified as triggered, further decoding is performed using the maximum likelihood criterion. Let $P[\varepsilon]$ denote the average error rate. In this proof, let ε_1 and ε_2 denote the errors in the two steps respectively. According to the law of total probability, we have

$$P[\varepsilon] = p(P[\varepsilon_1 \mid \mathcal{D}] + P[\varepsilon_2 \mid \bar{\varepsilon}_1, \mathcal{D}]P[\bar{\varepsilon}_1 \mid \mathcal{D}]) + (1 - p)P[\varepsilon \mid \bar{\mathcal{D}}]. \quad (99)$$

Next, we analyze the three terms on the right side of the equation separately.

For large blocklength n , there is a phenomenon called sphere hardening. Specifically, the received symbols may concentrate near the surface of the n -dimensional sphere.

According to Chapter 5.5 of [40], the mean and variance of $|\underline{z}|^2$ is given by

$$\mathbb{E}[|\underline{z}|^2] = 1, \quad \sigma^2(|\underline{z}|^2) = \frac{2}{n}. \quad (100)$$

It follows from Chebyshev's inequality that

$$\Pr[||\underline{z}|^2 - 1| \geq \Delta] \leq \frac{2}{n\Delta^2}, \quad \forall \Delta > 0, \quad (101)$$

which implies that as n becomes sufficiently large, the probability of $|\underline{z}|^2$ deviating from its average value of 1 by more than Δ approaches 0. By taking $\Delta = \frac{P}{2}$, we have

$$P[\varepsilon \mid \bar{\mathcal{D}}] = \Pr\left[|\underline{z}|^2 \geq \frac{P}{2} + 1\right] < \frac{8}{nP^2}. \quad (102)$$

Therefore, the third term in Eq. (99) satisfies $P[\varepsilon \mid \bar{\mathcal{D}}] < \frac{\epsilon}{3}$ for sufficiently large n .

We next consider the case where the transmitter is in a triggered state. When codeword $\underline{s}_k$ is transmitted, the mean and variance of $|\underline{y}|^2$ is given in [40] as

$$\mathbb{E}[|\underline{y}|^2] = |\underline{s}_k|^2 + 1, \quad \sigma^2(|\underline{y}|^2) \leq \frac{2}{n}(4P + 2). \quad (103)$$

With Chebyshev's inequality, we have

$$\begin{aligned} \Pr\{|\underline{y}|^2 \leq |\underline{s}_k|^2 + 1 - \Delta\} &< \Pr\{||\underline{y}|^2 - |\underline{s}_k|^2 - 1| \geq \Delta\} \\ &\leq \frac{2}{n\Delta^2}(4P + 2), \end{aligned} \quad (104)$$

which tends to zero as n gets large. Eq. (104) shares a similar form with Eq. (101). However, the radius $|\underline{s}_k|^2 + 1$ depends on the transmitted codeword, which poses challenges for subsequent analysis. To address this issue, we apply the following scaling technique:

$$\begin{aligned} &\Pr\{|\underline{y}|^2 \leq P + 1 - 2\Delta\} \\ &= \int_0^P \Pr\{|\underline{y}|^2 \leq P + 1 - 2\Delta + x - x \mid x\} p_{|\underline{s}_k|^2}(x) dx \\ &= \int_0^{P-\Delta} \Pr\{|\underline{y}|^2 \leq P + 1 - 2\Delta + x - x \mid x\} p_{|\underline{s}_k|^2}(x) dx \\ &\quad + \int_{P-\Delta}^P \Pr\{|\underline{y}|^2 \leq x + 1 - (x + 2\Delta - P) \mid x\} p_{|\underline{s}_k|^2}(x) dx \\ &\leq \int_0^{P-\Delta} p_{|\underline{s}_k|^2}(x) dx \\ &\quad + \int_{P-\Delta}^P \Pr\{|\underline{y}|^2 \leq x + 1 - \Delta \mid x\} p_{|\underline{s}_k|^2}(x) dx \\ &< \Pr\{|\underline{s}_i| \leq \sqrt{P - \Delta}\} \\ &\quad + \Pr\{|\underline{y}|^2 \leq |\underline{s}_k|^2 + 1 - \Delta\} \int_{P-\Delta}^P p_{|\underline{s}_k|^2}(x) dx \\ &< \Pr\{|\underline{s}_i| \leq \sqrt{P - \Delta}\} + \Pr\{|\underline{y}|^2 \leq |\underline{s}_k|^2 + 1 - \Delta\}. \end{aligned} \quad (105)$$

By substituting Eq. (98) and Eq. (104) into Eq. (105), we have

$$\Pr\{|\underline{y}|^2 \leq P + 1 - 2\Delta\} < \left(1 - \frac{\Delta}{P}\right)^{n/2} + \frac{2}{n\Delta^2}(4P + 2). \quad (106)$$

By taking $\Delta = \frac{P}{4}$, we have

$$\begin{aligned} P[\varepsilon_1 | \mathcal{D}] &= \Pr\left[|\underline{y}|^2 \leq \frac{P}{2} + 1 \mid \mathcal{D}\right] \\ &< \left(\frac{3}{4}\right)^{n/2} + \frac{32}{nP^2}(4P + 2). \end{aligned} \quad (107)$$

Therefore, the first term in Eq. (99) satisfies $P[\varepsilon_1 | \mathcal{D}] < \frac{\epsilon}{3}$ for sufficiently large n .

For the second term in Eq. (99), we have

$$P[\bar{\varepsilon}_1 | \mathcal{D}]P[\varepsilon_2 | \bar{\varepsilon}_1, \mathcal{D}] < P_{\text{conv}}[\varepsilon], \quad (109)$$

where $P_{\text{conv}}[\varepsilon]$ represents the average decoding error rate in conventional communication. According to the capacity theorem for AWGN channels, $P_{\text{conv}}[\varepsilon]$ can be made arbitrarily small for sufficiently large n if $R < C$, where $C = \frac{1}{2} \log(1 + \gamma)$ denotes the AWGN channel capacity.

Finally, by considering the upper bounds of all three terms in Eq. (99), we establish that $P[\varepsilon] < \epsilon$ if $R < C$ and n is sufficiently large. This demonstrates that $C_{\text{ETC}|\mathcal{D}} \geq C$, where $C_{\text{ETC}|\mathcal{D}}$ denotes the channel capacity in the triggered state of ETC. Additionally, in the triggered state, the equivalent codeword from the idle state, θ , introduces additional interference. Consequently, the channel capacity with codebook $\mathcal{C}'_n$ cannot exceed the capacity with codebook $\mathcal{C}_n$, implying $C_{\text{ETC}|\mathcal{D}} \leq C$. Therefore, we conclude that $C_{\text{ETC}|\mathcal{D}} = C$, completing the proof. It is important to note that codebooks satisfying equal-power constraint can be constructed similarly, though the specific proof process is not repeated here.

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